RANDOM FOREST (REGRESSION)

In [1]:
```
# IMPORTING LIBRARIES
import numpy as np
import pandas as pd
import matplotlib.pyplot as mlt
import warnings
warnings.filterwarnings("ignore")
```

In [2]:
```
#LOADING DATASET
from sklearn.datasets import fetch_california_housing
data=fetch_california_housing()
data
```

Out[2]:
{'data': array([[8.3252   ,   41.   ,   6.98412698, ...,   2.55555556,
           37.88   , -122.23   ],
        [8.3014   ,   21.   ,   6.23813708, ...,   2.10984183,
           37.86   , -122.22   ],
        [7.2574   ,   52.   ,   8.28813559, ...,   2.80225989,
           37.85   , -122.24   ],
        ...,
        [1.7   ,   17.   ,   5.20554273, ...,   2.3256351 ,
           39.43   , -121.22   ],
        [1.8672   ,   18.   ,   5.32951289, ...,   2.12320917,
           39.43   , -121.32   ],
        [2.3886   ,   16.   ,   5.25471698, ...,   2.61698113,
           39.37   , -121.24   ]], shape=(20640, 8)),
 'target': array([4.526, 3.585, 3.521, ..., 0.923, 0.847, 0.894], shape=(20640,)),
 'frame': None,
 'target_names': ['MedHouseVal'],
 'feature_names': ['MedInc',
  'HouseAge',
  'AveRooms',
  'AveBedrms',
  'Population',
  'AveOccup',
  'Latitude',
  'Longitude'],
 'DESCR': '.. _california_housing_dataset:\n\nCalifornia Housing dataset\n--------
------------------\n\n**Data Set Characteristics:**\n\n:Number of Instances: 20640
\n\n:Number of Attributes: 8 numeric, predictive attributes and the target\n\n:Att
ribute Information:\n   - MedInc   median income in block group\n   - House
Age   median house age in block group\n   - AveRooms   average number of ro
oms per household\n   - AveBedrms   average number of bedrooms per household\n
- Population   block group population\n   - AveOccup   average number of hous
ehold members\n   - Latitude   block group latitude\n   - Longitude   block
group longitude\n\n:Missing Attribute Values: None\n\nThis dataset was obtained fr
om the StatLib repository.\nhttps://www.dcc.fc.up.pt/~ltorgo/Regression/cal_housin
g.html\n\nThe target variable is the median house value for California district
s,\nexpressed in hundreds of thousands of dollars ($100,000).\n\nThis dataset was
derived from the 1990 U.S. census, using one row per census\nblock group. A block
group is the smallest geographical unit for which the U.S.\nCensus Bureau publishe
s sample data (a block group typically has a population\nof 600 to 3,000 peopl
e).\n\nA household is a group of people residing within a home. Since the average
\nnumber of rooms and bedrooms in this dataset are provided per household, these\n
columns may take surprisingly large values for block groups with few households\na
nd many empty houses, such as vacation resorts.\n\nIt can be downloaded/loaded usi
ng the\n:func:`sklearn.datasets.fetch_california_housing` function.\n\n.. rubric::
References\n\n- Pace, R. Kelley and Ronald Barry, Sparse Spatial Autoregression
s,\n Statistics and Probability Letters, 33:291-297, 1997.\n'}

In [3]:
```
#CONVERTING DATA INTO A DATA FRAME
df=pd.DataFrame(np.c_[data.data,data.target],
                columns=[list(data.feature_names)+["target"]])
df
```

Out[3]:

	MedInc	HouseAge	AveRooms	AveBedrms	Population	AveOccup	Latitude	Long
0	8.3252	41.0	6.984127	1.023810	322.0	2.555556	37.88	-1
1	8.3014	21.0	6.238137	0.971880	2401.0	2.109842	37.86	-1
2	7.2574	52.0	8.288136	1.073446	496.0	2.802260	37.85	-1
3	5.6431	52.0	5.817352	1.073059	558.0	2.547945	37.85	-1
4	3.8462	52.0	6.281853	1.081081	565.0	2.181467	37.85	-1
...	...	...	...	...	...	...	...	
20635	1.5603	25.0	5.045455	1.133333	845.0	2.560606	39.48	-1
20636	2.5568	18.0	6.114035	1.315789	356.0	3.122807	39.49	-1
20637	1.7000	17.0	5.205543	1.120092	1007.0	2.325635	39.43	-1
20638	1.8672	18.0	5.329513	1.171920	741.0	2.123209	39.43	-1
20639	2.3886	16.0	5.254717	1.162264	1387.0	2.616981	39.37	-1

20640 rows × 9 columns

In [4]:
```
# RANDOM FOREST (REGRESSION)
# STEP 1: ASSIGNING VARIABLES(INDEPENDENT/DEPENDENT)

# INDEPENDENT VARIABLE
x=df.drop("target",axis=1)

# DEPENDENT VARIABLE
y=df[["target"]]
```

In [5]:
```
# STEP 2:IMPORT TRAIN TEST SPLIT
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test,=train_test_split(x,y,test_size=0.3,random_state=42)
```

In [6]:
```
print("Original size : ",df.shape)
print("x_train size : ",x_train.shape)
print("x_test size : ",x_test.shape)
print("y_train size : ",y_train.shape)
print("y_test size : ",y_test.shape)
```

```
Original size :  (20640, 9)
x_train size :  (14448, 8)
x_test size :  (6192, 8)
y_train size :  (14448, 1)
y_test size :  (6192, 1)
```

In [7]:
```
# STEP 3: APPLY RANDOM FOREST CLASSIFICATION
from sklearn.ensemble import RandomForestRegressor
regressor=RandomForestRegressor(n_estimators=50)
regressor
```

Out[7]:
```
RandomForestRegressor
▶ Parameters
```

In [8]:
```
# STEP 4 :FIT TO DATA
# Use only train(x_train,y_train)
regressor.fit(x_train,y_train)
```

Out[8]:
```
RandomForestRegressor
▶ Parameters
```

In [9]:
```
# CHECK ACCURACY
regressor.score(x_test,y_test)
```

Out[9]:
```
0.8056665377222805
```

In [10]:
```
# PREDICTION OF NEW DATA
# IMPORT NUMPY FIRST(DATA MUST IN ARRAY FORM)
house=np.array([[8.3014,21.0,6.238137,0.971880,2401.0,2.109842,37.86,-122.22]])
house
```

Out[10]:
```
array([[ 8.301400e+00,  2.100000e+01,  6.238137e+00,  9.718800e-01,
         2.401000e+03,  2.109842e+00,  3.786000e+01, -1.222200e+02]])
```

In [11]:
```
prediction=regressor.predict(house)
prediction
```

Out[11]:
```
array([4.0032622])
```

In [12]:
```
# EVALUATE MODEL
from sklearn.metrics import(mean_absolute_error,mean_squared_error,r2_score)
```

In [13]:
```
y_pred=regressor.predict(x_test)
mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)

print("Mean Absolute Error:", mae)
print("Mean Squared Error:", mse)
print("Root Mean Squared Error:", rmse)
print("R2 Score:", r2)
```

```
Mean Absolute Error: 0.3320465444767442
Mean Squared Error: 0.2550704635298394
Root Mean Squared Error: 0.5050450113899151
R2 Score: 0.8056665377222805
```